No.	Status	Date
LNDE-RPT-01-037-DS-6-SG-101	Issued	2001-03-02

UT Thickness Data Sheet - 6-SG-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 6-SG-101 located within the ATU-100. The scope of work is defined by the 5-year Class 1 survey of sour gas and rich amine piping as detailed in LNDE-RPT-01-037.

Grace Okafor, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., is responsible for the execution of the thickness measurements and the recording of the data presented herein.

Technical Basis

All measurements were performed under the requirements of QSCP-NDT-001 Rev 0, the site NDE procedure for ultrasonic thickness measurement at ATU-100. The inspection methodology conforms to the standard ASTM E797.

The line 6-SG-101 is constructed from A106 Gr.B with an outer diameter of 168.3 mm and a nominal thickness of 7.11 mm. Design limits for this line are 1965 kPa(g) and 66 °C.

Acceptance Criteria

The retirement thickness for 6-SG-101 is established as the t-min of 3.56 mm. Any reading recorded below this t-min value shall be raised as an NCR.

Corrosion rates are determined by the larger of the long-term rate or the short-term rate over the latest pair of readings at least 3 years apart. Remaining life is calculated as (t - t-min) / rate.

Inspection Interval

The next thickness reading is due at no more than half the remaining life, subject to the class maximum. As a Class 1 line (sour gas, regenerator overhead), the maximum interval for thickness and external visual inspection is 5 years.

The current data is integrated into the overall survey findings reported in LNDE-RPT-01-037.

Thickness readings, measured 2001-03-02

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	1996-03-08	7.12	6.82	0.060	0.060	>50	2006-03-02	
6-SG-101-02	7.11	3.56	1996-03-08	7.05	6.71	0.068	0.068	46.2	2006-03-02	
6-SG-101-03	7.11	3.56	1996-03-08	7.08	6.74	0.068	0.068	46.6	2006-03-02	
6-SG-101-04	7.11	3.56	1996-03-08	7.09	6.73	0.072	0.072	43.9	2006-03-02	
6-SG-101-05	7.11	3.56	1996-03-08	7.14	6.68	0.092	0.092	33.8	2006-03-02	
6-SG-101-06	7.11	3.56	1996-03-08	7.07	6.87	0.040	0.040	>50	2006-03-02	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	1996-03-08	7.14	6.72	0.084	0.084	37.5	2006-03-02	
6-SG-102-02	7.11	3.56	1996-03-08	7.11	6.72	0.078	0.078	40.4	2006-03-02	
6-SG-102-03	7.11	3.56	1996-03-08	7.04	6.69	0.070	0.070	44.6	2006-03-02	
6-SG-102-04	7.11	3.56	1996-03-08	6.99	6.64	0.070	0.070	43.8	2006-03-02	
6-SG-102-05	7.11	3.56	1996-03-08	7.13	6.93	0.040	0.040	>50	2006-03-02	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	1996-03-08	8.52	8.39	0.026	0.026	>50	2011-03-02	
4-RA-101-02	8.56	3.81	1996-03-08	8.54	8.45	0.018	0.018	>50	2011-03-02	
4-RA-101-03	8.56	3.81	1996-03-08	8.61	8.42	0.038	0.038	>50	2011-03-02	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-04	8.56	3.81	1996-03-08	8.63	8.37	0.052	0.052	>50	2011-03-02	
4-RA-101-05	8.56	3.81	1996-03-08	8.60	8.42	0.036	0.036	>50	2011-03-02	
4-RA-101-07	8.56	3.81	1996-03-08	8.60	8.51	0.018	0.018	>50	2011-03-02	
4-RA-101-08	8.56	3.81	1996-03-08	8.68	8.30	0.076	0.076	>50	2011-03-02	
4-RA-101-09	8.56	3.81	1996-03-08	8.57	8.43	0.028	0.028	>50	2011-03-02	
4-RA-101-10	8.56	3.81	1996-03-08	8.67	8.48	0.038	0.038	>50	2011-03-02	
4-RA-101-11	8.56	3.81	1996-03-08	8.68	8.39	0.058	0.058	>50	2011-03-02	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	1996-03-08	8.54	8.48	0.012	0.012	>50	2011-03-02	
4-RA-102-02	8.56	3.81	1996-03-08	8.57	8.50	0.014	0.014	>50	2011-03-02	
4-RA-102-03	8.56	3.81	1996-03-08	8.56	8.37	0.038	0.038	>50	2011-03-02	
4-RA-102-04	8.56	3.81	1996-03-08	8.54	8.35	0.038	0.038	>50	2011-03-02	
4-RA-102-05	8.56	3.81	1996-03-08	8.55	8.36	0.038	0.038	>50	2011-03-02	
4-RA-102-06	8.56	3.81	1996-03-08	8.51	8.38	0.026	0.026	>50	2011-03-02	
4-RA-102-07	8.56	3.81	1996-03-08	8.58	8.49	0.018	0.018	>50	2011-03-02	
4-RA-102-08	8.56	3.81	1996-03-08	8.49	8.36	0.026	0.026	>50	2011-03-02	
4-RA-102-09	8.56	3.81	1996-03-08	8.55	8.35	0.040	0.040	>50	2011-03-02	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	1996-03-08	11.00	10.82	0.036	0.036	>50	2011-03-02	
6-RA-103-02	10.97	4.57	1996-03-08	11.09	10.69	0.080	0.080	>50	2011-03-02	
6-RA-103-03	10.97	4.57	1996-03-08	10.99	10.84	0.030	0.030	>50	2011-03-02	
6-RA-103-04	10.97	4.57	1996-03-08	10.98	10.59	0.078	0.078	>50	2011-03-02	
6-RA-103-05	10.97	4.57	1996-03-08	10.95	10.61	0.068	0.068	>50	2011-03-02	
6-RA-103-06	10.97	4.57	1996-03-08	11.01	10.86	0.030	0.030	>50	2011-03-02	
6-RA-103-07	10.97	4.57	1996-03-08	10.99	10.79	0.040	0.040	>50	2011-03-02	
6-RA-103-08	10.97	4.57	1996-03-08	10.97	10.83	0.028	0.028	>50	2011-03-02	
6-RA-103-09	10.97	4.57	1996-03-08	11.01	10.85	0.032	0.032	>50	2011-03-02	
6-RA-103-10	10.97	4.57	1996-03-08	11.02	10.78	0.048	0.048	>50	2011-03-02	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	1996-03-08	7.59	7.26	0.066	0.066	>50	2006-03-02	
3-RV-101-02	7.62	3.56	1996-03-08	7.57	7.31	0.052	0.052	>50	2006-03-02	
3-RV-101-03	7.62	3.56	1996-03-08	7.52	7.15	0.074	0.074	48.3	2006-03-02	
3-RV-101-04	7.62	3.56	1996-03-08	7.59	7.18	0.082	0.082	44.0	2006-03-02	
3-RV-101-05	7.62	3.56	1996-03-08	7.67	7.25	0.084	0.084	43.8	2006-03-02	
3-RV-101-06	7.62	3.56	1996-03-08	7.59	7.20	0.078	0.078	46.5	2006-03-02	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	1996-03-08	5.53	5.29	0.048	0.048	>50	2006-03-02	
2-RV-102-02	5.54	2.79	1996-03-08	5.59	5.12	0.094	0.094	24.7	2006-03-02	
2-RV-102-03	5.54	2.79	1996-03-08	5.48	5.24	0.048	0.048	>50	2006-03-02	
2-RV-102-04	5.54	2.79	1996-03-08	5.58	5.33	0.050	0.050	>50	2006-03-02	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	1996-03-08	8.51	8.40	0.022	0.022	>50	2011-03-02	
4-RA-104-02	8.56	3.81	1996-03-08	8.55	8.41	0.028	0.028	>50	2011-03-02	
4-RA-104-03	8.56	3.81	1996-03-08	8.54	8.46	0.016	0.016	>50	2011-03-02	
4-RA-104-04	8.56	3.81	1996-03-08	8.59	8.35	0.048	0.048	>50	2011-03-02	
4-RA-104-05	8.56	3.81	1996-03-08	8.60	8.29	0.062	0.062	>50	2011-03-02	
4-RA-104-06	8.56	3.81	1996-03-08	8.59	8.39	0.040	0.040	>50	2011-03-02	
4-RA-104-07	8.56	3.81	1996-03-08	8.59	8.29	0.060	0.060	>50	2011-03-02	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	2001-03-02	_____